

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



## Klübersynth AG 14-61

|         |                |                                 |             |
|---------|----------------|---------------------------------|-------------|
| Version | Revision Date: | Date of last issue: 07.12.2020  | Print Date: |
| 3.7     | 06.07.2021     | Date of first issue: 17.05.2013 | 06.07.2021  |

### SECTION 1: Identification of the substance/mixture and of the company/undertaking

#### 1.1 Product identifier

Product name : Klübersynth AG 14-61

Article-No. : 017184

#### 1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub-  
stance/Mixture : Grease

Recommended restrictions  
on use : Restricted to professional users.

#### 1.3 Details of the supplier of the safety data sheet

Company : Klüber Lubrication München  
Geisenhausenerstr. 7  
81379 München  
Deutschland  
Tel: +49 (0) 89 7876 0  
Fax: +49 (0) 89 7876 333  
info@klueber.com

E-mail address of person  
responsible for the SDS : mcm@klueber.com  
Material Compliance Management

National contact : Klüber Lubrication Great Britain Limited  
Unit 10 Pennine Business Park  
Longbow Close  
Huddersfield  
West Yorkshire HD2 1GQ  
Great Britain  
Tel: +44-1422-205115  
Fax: +44-1422-206073  
sales@uk.klueber.com

#### 1.4 Emergency telephone number

Emergency telephone num-  
ber : +49 89 7876 700 (24 hrs)

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### SECTION 2: Hazards identification

#### 2.1 Classification of the substance or mixture

##### Classification (REGULATION (EC) No 1272/2008)

Not a hazardous substance or mixture.

#### 2.2 Label elements

##### Labelling (REGULATION (EC) No 1272/2008)

Not a hazardous substance or mixture.

##### Additional Labelling

EUH210 Safety data sheet available on request.

#### 2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

### SECTION 3: Composition/information on ingredients

#### 3.2 Mixtures

Chemical nature : Synthetic hydrocarbon oil  
solid lubricant  
aluminium complex soap

##### Components

| Chemical name                                                                                                                                                                   | CAS-No.<br>EC-No.<br><br>Index-No.<br>Registration number | Classification                                                    | Concentration<br>limits<br>M-Factor<br>Notes | Concentration<br>(% w/w) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------|--------------------------|
| 1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol                                                                             | 939-692-2                                                 | Aquatic Chronic3;<br>H412                                         |                                              | >= 1 - < 2.5             |
| Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylene-poly-, triethylenetetramine fraction and 3-(C9-C15, C12 rich, alk-1-enyl)dihydro-2,5-furandione | 947-263-6                                                 | Skin Irrit.2; H315<br>Repr.2; H361fd<br>Aquatic Chronic4;<br>H413 |                                              | >= 1 - < 2.5             |
| Substances with a workplace exposure limit :                                                                                                                                    |                                                           |                                                                   |                                              |                          |

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|                 |                        |                |  |             |
|-----------------|------------------------|----------------|--|-------------|
| magnesium oxide | 1309-48-4<br>215-171-9 | Not classified |  | >= 1 - < 10 |
|-----------------|------------------------|----------------|--|-------------|

For explanation of abbreviations see section 16.

## SECTION 4: First aid measures

### 4.1 Description of first aid measures

- If inhaled : Obtain medical attention.  
Remove person to fresh air. If signs/symptoms continue, get medical attention.  
Keep patient warm and at rest.  
If unconscious, place in recovery position and seek medical advice.  
Keep respiratory tract clear.  
If breathing is irregular or stopped, administer artificial respiration.
- In case of skin contact : Take off all contaminated clothing immediately.  
Get medical attention immediately if irritation develops and persists.  
Wash clothing before reuse.  
Thoroughly clean shoes before reuse.  
Wash off immediately with plenty of water.
- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 10 minutes.  
If eye irritation persists, consult a specialist.
- If swallowed : Move the victim to fresh air.  
If unconscious, place in recovery position and seek medical advice.  
Keep respiratory tract clear.  
Do not induce vomiting without medical advice.  
Obtain medical attention.  
Never give anything by mouth to an unconscious person.

### 4.2 Most important symptoms and effects, both acute and delayed

- Symptoms : No information available.
- Risks : None known.

### 4.3 Indication of any immediate medical attention and special treatment needed

- Treatment : No information available.

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### SECTION 5: Firefighting measures

#### 5.1 Extinguishing media

Suitable extinguishing media : Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Unsuitable extinguishing media : High volume water jet

#### 5.2 Special hazards arising from the substance or mixture

Hazardous combustion products : Carbon oxides  
Nitrogen oxides (NO<sub>x</sub>)  
Sulphur oxides  
Metal oxides

#### 5.3 Advice for firefighters

Special protective equipment for firefighters : In the event of fire, wear self-contained breathing apparatus. Use personal protective equipment. Exposure to decomposition products may be a hazard to health.

Further information : Standard procedure for chemical fires.

### SECTION 6: Accidental release measures

#### 6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Evacuate personnel to safe areas.  
Use the indicated respiratory protection if the occupational exposure limit is exceeded and/or in case of product release (dust).  
Do not breathe vapours, aerosols.  
Refer to protective measures listed in sections 7 and 8.

#### 6.2 Environmental precautions

Environmental precautions : Try to prevent the material from entering drains or water courses.  
Local authorities should be advised if significant spillages cannot be contained.

#### 6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Clean up promptly by sweeping or vacuum.  
Keep in suitable, closed containers for disposal.

#### 6.4 Reference to other sections

For personal protection see section 8.

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### SECTION 7: Handling and storage

#### 7.1 Precautions for safe handling

Advice on safe handling : Avoid contact with skin and eyes.  
For personal protection see section 8.  
Smoking, eating and drinking should be prohibited in the application area.  
Wash hands and face before breaks and immediately after handling the product.  
Do not get in eyes or mouth or on skin.  
Do not get on skin or clothing.  
Do not ingest.  
Do not repack.  
These safety instructions also apply to empty packaging which may still contain product residues.  
Keep container closed when not in use.

Hygiene measures : Wash face, hands and any exposed skin thoroughly after handling.

#### 7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Store in original container. Keep container closed when not in use. Keep in a dry, cool and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Store in accordance with the particular national regulations. Keep in properly labelled containers.

#### 7.3 Specific end use(s)

Specific use(s) : Specific instructions for handling, not required.

### SECTION 8: Exposure controls/personal protection

#### 8.1 Control parameters

##### Occupational Exposure Limits

| Components      | CAS-No.   | Value type (Form of exposure) | Control parameters               | Basis                |
|-----------------|-----------|-------------------------------|----------------------------------|----------------------|
| magnesium oxide | 1309-48-4 | TWA (inhalable dust)          | 10 mg/m <sup>3</sup> (Magnesium) | GB EH40 (2020-01-01) |
|                 |           | TWA (Respirable dust)         | 4 mg/m <sup>3</sup> (Magnesium)  | GB EH40 (2020-01-01) |
|                 |           | TWA (Fumes)                   | 10 mg/m <sup>3</sup> (Magnesium) | GB EH40 (2020-01-01) |

##### Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006:

| Substance name | End Use | Exposure routes | Potential health effects | Value |
|----------------|---------|-----------------|--------------------------|-------|
|----------------|---------|-----------------|--------------------------|-------|

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|                                                                                                                                                                                |         |              |                            |                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------|----------------------------|-------------------|
| zinc sulphide                                                                                                                                                                  | Workers | Inhalation   | Long-term systemic effects | 5 mg/m3           |
|                                                                                                                                                                                | Workers | Dermal       | Long-term systemic effects | 83 mg/kg bw/day   |
| 1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol                                                                            | Workers | Inhalation   | Long-term systemic effects | 4.408 mg/m3       |
|                                                                                                                                                                                | Workers | Skin contact | Long-term systemic effects | 6.25 mg/kg bw/day |
| Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethylenetetramine fraction and 3-(C9-C15, C12 rich, alk-1-enyl)dihydro-2,5-furandione | Workers | Inhalation   | Long-term systemic effects | 3.72 mg/m3        |
|                                                                                                                                                                                | Workers | Skin contact | Long-term systemic effects | 1.04 mg/m3        |

### Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006:

| Substance name                                                                                                             | Environmental Compartment                            | Value        |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|--------------|
| zinc sulphide                                                                                                              | Fresh water                                          | 0.0206 mg/l  |
|                                                                                                                            | Marine water                                         | 0.0061 mg/l  |
|                                                                                                                            | Microbiological Activity in Sewage Treatment Systems | 0.1 mg/l     |
|                                                                                                                            | Fresh water sediment                                 | 117.8 mg/kg  |
|                                                                                                                            | Marine sediment                                      | 56.5 mg/kg   |
|                                                                                                                            | Soil                                                 | 35.6 mg/kg   |
| distillates (petroleum), solvent-dewaxed heavy paraffinic                                                                  | Oral                                                 | 9.33 mg/kg   |
| 1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol                        | Fresh water                                          | 0.041 mg/l   |
|                                                                                                                            | Marine water                                         | 0.0041 mg/l  |
|                                                                                                                            | Intermittent use/release                             | 0.41 mg/l    |
|                                                                                                                            | Microbiological Activity in Sewage Treatment Systems | 8000 mg/l    |
|                                                                                                                            | Fresh water sediment                                 | 380.62 mg/kg |
|                                                                                                                            | Marine sediment                                      | 38.06 mg/kg  |
|                                                                                                                            | Soil                                                 | 308.98 mg/kg |
|                                                                                                                            | Oral                                                 | 6.67 mg/kg   |
| Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethylenetetramine fraction and 3- | Fresh water                                          | 0.496 mg/l   |

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|                                                      |                                                      |                  |
|------------------------------------------------------|------------------------------------------------------|------------------|
| (C9–C15, C12 rich, alk-1-enyl)dihydro-2,5-furandione |                                                      |                  |
|                                                      | Marine water                                         | 0.05 mg/l        |
|                                                      | Microbiological Activity in Sewage Treatment Systems | 100 mg/l         |
|                                                      | Fresh water sediment                                 | 3772830.55 mg/kg |
|                                                      | Marine sediment                                      | 377283.06 mg/kg  |
|                                                      | Soil                                                 | 3935351.65 mg/kg |

### 8.2 Exposure controls

#### Engineering measures

Handle only in a place equipped with local exhaust (or other appropriate exhaust).

#### Personal protective equipment

Eye protection : Safety glasses with side-shields

Hand protection

Material : Nitrile rubber

Break through time : > 10 min

Protective index : Class 1

Remarks : For prolonged or repeated contact use protective gloves. The break through time depends amongst other things on the material, the thickness and the type of glove and therefore has to be measured for each case.  
The selected protective gloves have to satisfy the specifications of Regulation (EU) 2016/425 and the standard EN 374 derived from it.

Respiratory protection : Not required; except in case of aerosol formation.

Filter type : Filter type P

Protective measures : The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.  
Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place.

## SECTION 9: Physical and chemical properties

### 9.1 Information on basic physical and chemical properties

Appearance : paste

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|                                                  |   |                                                                       |
|--------------------------------------------------|---|-----------------------------------------------------------------------|
| Colour                                           | : | white                                                                 |
| Odour                                            | : | characteristic                                                        |
| Odour Threshold                                  | : | No data available                                                     |
| pH                                               | : | Not applicable                                                        |
| Melting point/range                              | : | No data available                                                     |
| Boiling point/boiling range                      | : | No data available                                                     |
| Flash point                                      | : | Not applicable                                                        |
| Evaporation rate                                 | : | No data available                                                     |
| Flammability (solid, gas)                        | : | Combustible Solids                                                    |
| Upper explosion limit / Upper flammability limit | : | No data available                                                     |
| Lower explosion limit / Lower flammability limit | : | No data available                                                     |
| Vapour pressure                                  | : | < 0.001 hPa (20 °C)                                                   |
| Relative vapour density                          | : | No data available                                                     |
| Relative density                                 | : | 1.00 (20 °C)<br>Reference substance: Water<br>The value is calculated |
| Density                                          | : | 1.00 g/cm <sup>3</sup><br>(20 °C)                                     |
| Bulk density                                     | : | No data available                                                     |
| Solubility(ies)                                  |   |                                                                       |
| Water solubility                                 | : | insoluble                                                             |
| Solubility in other solvents                     | : | No data available                                                     |
| Partition coefficient: n-octanol/water           | : | No data available                                                     |
| Auto-ignition temperature                        | : | No data available                                                     |
| Decomposition temperature                        | : | No data available                                                     |
| Viscosity                                        |   |                                                                       |



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Viscosity, dynamic : No data available

Viscosity, kinematic : Not applicable

Explosive properties : Not explosive

Oxidizing properties : No data available

### 9.2 Other information

Sublimation point : No data available

Self-ignition : No data available

## SECTION 10: Stability and reactivity

### 10.1 Reactivity

No hazards to be specially mentioned.

### 10.2 Chemical stability

Stable under normal conditions.

### 10.3 Possibility of hazardous reactions

Hazardous reactions : No dangerous reaction known under conditions of normal use.

### 10.4 Conditions to avoid

Conditions to avoid : No conditions to be specially mentioned.

### 10.5 Incompatible materials

Materials to avoid : No materials to be especially mentioned.

### 10.6 Hazardous decomposition products

No decomposition if stored and applied as directed.

## SECTION 11: Toxicological information

### 11.1 Information on toxicological effects

#### Acute toxicity

#### Product:

Acute inhalation toxicity : Remarks: This information is not available.

Acute dermal toxicity : Remarks: This information is not available.

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### Components:

#### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Acute oral toxicity : LD50 (Rat): > 5,000 mg/kg  
Method: OECD Test Guideline 401

Acute inhalation toxicity : LC50 (Rat): > 2.75 mg/l  
Exposure time: 4 h  
Test atmosphere: vapour  
Method: OECD Test Guideline 403  
Assessment: The substance or mixture has no acute inhalation toxicity  
Remarks: An LC50/inhalation/4h/rat could not be determined because no mortality of rats was observed at the maximum achievable concentration.  
Information given is based on data obtained from similar substances.

Acute dermal toxicity : LD50 (Rat): > 2,000 mg/kg  
Method: OECD Test Guideline 402  
Assessment: The substance or mixture has no acute dermal toxicity  
Remarks: Information given is based on data obtained from similar substances.

Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethylenetetramine fraction and 3-(C9-C15, C12 rich, alk-1-enyl) dihydro-2,5-furandione

:

Acute oral toxicity : LD50 (Rat): > 2,000 mg/kg  
Method: OECD Test Guideline 423  
GLP: yes

### **magnesium oxide:**

Acute oral toxicity : LD50 (Rat, male): 3,870 mg/kg

### **Skin corrosion/irritation**

### Product:

Remarks : This information is not available.

### Components:

#### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Species : Rabbit  
Assessment : No skin irritation  
Method : OECD Test Guideline 404  
Result : No skin irritation

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Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethylenetetramine fraction and 3-(C9–C15, C12 rich, alk-1-enyl)dihydro-2,5-furandione

:

|            |   |                         |
|------------|---|-------------------------|
| Species    | : | human skin              |
| Assessment | : | Irritating to skin.     |
| Method     | : | OECD Test Guideline 439 |
| Result     | : | Irritating to skin.     |
| GLP        | : | yes                     |

### Serious eye damage/eye irritation

#### Product:

Remarks : This information is not available.

#### Components:

##### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

|            |   |                         |
|------------|---|-------------------------|
| Species    | : | Rabbit                  |
| Assessment | : | No eye irritation       |
| Method     | : | OECD Test Guideline 405 |
| Result     | : | No eye irritation       |

Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethylenetetramine fraction and 3-(C9–C15, C12 rich, alk-1-enyl)dihydro-2,5-furandione

:

|            |   |                   |
|------------|---|-------------------|
| Assessment | : | No eye irritation |
| Result     | : | No eye irritation |

### Respiratory or skin sensitisation

#### Product:

Remarks : This information is not available.

#### Components:

##### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

|            |   |                                                    |
|------------|---|----------------------------------------------------|
| Test Type  | : | Buehler Test                                       |
| Species    | : | Guinea pig                                         |
| Assessment | : | Did not cause sensitisation on laboratory animals. |
| Method     | : | OECD Test Guideline 406                            |
| Result     | : | Did not cause sensitisation on laboratory animals. |

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:

|            |   |                                                    |
|------------|---|----------------------------------------------------|
| Assessment | : | Did not cause sensitisation on laboratory animals. |
| Result     | : | Did not cause sensitisation on laboratory animals. |

### Germ cell mutagenicity

#### Product:

Genotoxicity in vitro : Remarks: No data available

Genotoxicity in vivo : Remarks: No data available

#### Components:

##### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test  
Test system: Chinese hamster fibroblasts  
Metabolic activation: with and without metabolic activation  
Method: OECD Test Guideline 473  
Result: negative  
Remarks: Information given is based on data obtained from similar substances.

Germ cell mutagenicity- Assessment : Tests on bacterial or mammalian cell cultures did not show mutagenic effects.

### Carcinogenicity

#### Product:

Remarks : No data available

#### Components:

##### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Carcinogenicity - Assessment : Not classifiable as a human carcinogen.

### Reproductive toxicity

#### Product:

Effects on fertility : Remarks: No data available

Effects on foetal development : Remarks: No data available

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### Components:

#### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Effects on fertility : Species: Rat  
Application Route: Oral  
General Toxicity - Parent: NOAEL: 1,000 mg/kg body weight  
General Toxicity F1: NOAEL: 1,000 mg/kg body weight  
Method: OECD Test Guideline 421  
Remarks: Information given is based on data obtained from similar substances.

Reproductive toxicity - Assessment : No toxicity to reproduction  
Animal testing did not show any effects on foetal development.

Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethylenetetramine fraction and 3-(C9-C15, C12 rich, alk-1-enyl) dihydro-2,5-furandione

:  
Reproductive toxicity - Assessment : Some evidence of adverse effects on development, based on animal experiments., Some evidence of adverse effects on sexual function and fertility, based on animal experiments.  
Some evidence of adverse effects on development, based on animal experiments., Some evidence of adverse effects on sexual function and fertility, based on animal experiments.

### **STOT - single exposure**

#### Components:

#### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

### **STOT - repeated exposure**

#### Components:

#### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

### **Repeated dose toxicity**

#### Product:

Remarks : This information is not available.

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### Components:

#### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

|                   |   |                                                                      |
|-------------------|---|----------------------------------------------------------------------|
| Species           | : | Rat                                                                  |
| NOAEL             | : | 250 mg/kg                                                            |
| Application Route | : | Oral                                                                 |
| Method            | : | OECD Test Guideline 421                                              |
| Remarks           | : | Information given is based on data obtained from similar substances. |

### **Aspiration toxicity**

#### Product:

This information is not available.

### **Further information**

#### Product:

|         |   |                                                                                              |
|---------|---|----------------------------------------------------------------------------------------------|
| Remarks | : | Information given is based on data on the components and the toxicology of similar products. |
|---------|---|----------------------------------------------------------------------------------------------|

## SECTION 12: Ecological information

### 12.1 Toxicity

#### Product:

|                  |   |                            |
|------------------|---|----------------------------|
| Toxicity to fish | : | Remarks: No data available |
|------------------|---|----------------------------|

|                                                     |   |                            |
|-----------------------------------------------------|---|----------------------------|
| Toxicity to daphnia and other aquatic invertebrates | : | Remarks: No data available |
|-----------------------------------------------------|---|----------------------------|

|                                  |   |                            |
|----------------------------------|---|----------------------------|
| Toxicity to algae/aquatic plants | : | Remarks: No data available |
|----------------------------------|---|----------------------------|

|                            |   |                            |
|----------------------------|---|----------------------------|
| Toxicity to microorganisms | : | Remarks: No data available |
|----------------------------|---|----------------------------|

### Components:

#### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

|                  |   |                                                                                  |
|------------------|---|----------------------------------------------------------------------------------|
| Toxicity to fish | : | LC50 (Pimephales promelas (fathead minnow)): > 1,000 mg/l<br>Exposure time: 96 h |
|------------------|---|----------------------------------------------------------------------------------|

|                                                     |   |                                                                                                      |
|-----------------------------------------------------|---|------------------------------------------------------------------------------------------------------|
| Toxicity to daphnia and other aquatic invertebrates | : | EC50 (Daphnia magna (Water flea)): 41 mg/l<br>Exposure time: 48 h<br>Method: OECD Test Guideline 202 |
|-----------------------------------------------------|---|------------------------------------------------------------------------------------------------------|

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Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): > 100 mg/l  
Exposure time: 72 h  
Test Type: Growth inhibition  
Method: OECD Test Guideline 201

### Ecotoxicology Assessment

Acute aquatic toxicity : Harmful to aquatic life.

Chronic aquatic toxicity : Harmful to aquatic life with long lasting effects.

Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethylenetetramine fraction and 3-(C9–C15, C12 rich, alk-1-enyl)dihydro-2,5-furandione

:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): > 1,000 mg/l  
Exposure time: 96 h  
Test Type: static test  
Method: OECD Test Guideline 203  
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 1,000 mg/l  
Exposure time: 48 h  
Test Type: static test  
Method: OECD Test Guideline 202  
GLP: yes

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): 496 mg/l  
Exposure time: 72 h  
Test Type: static test  
Method: OECD Test Guideline 201  
GLP: yes

## 12.2 Persistence and degradability

### Product:

Biodegradability : Remarks: No data available

Physico-chemical removability : Remarks: No data available

### Components:

**1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Biodegradability : Test Type: Primary biodegradation  
Inoculum: activated sludge  
Result: Not readily biodegradable.  
Biodegradation: 0 %

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Exposure time: 28 d  
Method: OECD Test Guideline 301C

Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethyl-enetetramine fraction and 3-(C9–C15, C12 rich, alk-1-enyl)dihydro-2,5-furandione

:

Biodegradability : Test Type: anaerobic  
Inoculum: activated sludge  
Concentration: 3.77 mg/l  
Result: Not readily biodegradable.  
Biodegradation: 10 %  
Exposure time: 28 d  
Method: OECD Test Guideline 301D  
GLP: yes

### 12.3 Bioaccumulative potential

#### Product:

Bioaccumulation : Remarks: This mixture contains no substance considered to be persistent, bioaccumulating and toxic (PBT).  
This mixture contains no substance considered to be very persistent and very bioaccumulating (vPvB).

#### Components:

##### **1,3,4-Thiadiazolidine-2,5-dithione, reaction products with hydrogen peroxide and tert-dodecanethiol:**

Bioaccumulation : Species: Fish  
Bioconcentration factor (BCF): 3.16

Partition coefficient: n-octanol/water : log Pow: 8 (20 °C)

Reaction products of fatty acids, C16-18, C18 unsatd. with Amines, polyethylenepoly-, triethyl-enetetramine fraction and 3-(C9–C15, C12 rich, alk-1-enyl)dihydro-2,5-furandione

:

Partition coefficient: n-octanol/water : log Pow: > 10

#### **magnesium oxide:**

Partition coefficient: n-octanol/water : Remarks: Not applicable

### 12.4 Mobility in soil

#### Product:

Mobility : Remarks: No data available



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Distribution among environmental compartments : Remarks: No data available

### 12.5 Results of PBT and vPvB assessment

#### Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher..

### 12.6 Other adverse effects

#### Product:

Endocrine disrupting potential : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Additional ecological information : No information on ecology is available.

## SECTION 13: Disposal considerations

### 13.1 Waste treatment methods

Product : The product should not be allowed to enter drains, water courses or the soil.  
Do not dispose of with domestic refuse.  
Dispose of as hazardous waste in compliance with local and national regulations.

Waste codes should be assigned by the user based on the application for which the product was used.

Contaminated packaging : Packaging that is not properly emptied must be disposed of as the unused product.  
Dispose of waste product or used containers according to local regulations.

The following Waste Codes are only suggestions:

Waste Code : used product, unused product  
12 01 12\*, spent waxes and fats  
  
uncleaned packagings  
15 01 10, packaging containing residues of or contaminated by hazardous substances

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### SECTION 14: Transport information

#### 14.1 UN number

|      |                                     |
|------|-------------------------------------|
| ADR  | : Not regulated as a dangerous good |
| RID  | : Not regulated as a dangerous good |
| IMDG | : Not regulated as a dangerous good |
| IATA | : Not regulated as a dangerous good |

#### 14.2 UN proper shipping name

|      |                                     |
|------|-------------------------------------|
| ADR  | : Not regulated as a dangerous good |
| RID  | : Not regulated as a dangerous good |
| IMDG | : Not regulated as a dangerous good |
| IATA | : Not regulated as a dangerous good |

#### 14.3 Transport hazard class(es)

|      |                                     |
|------|-------------------------------------|
| ADR  | : Not regulated as a dangerous good |
| RID  | : Not regulated as a dangerous good |
| IMDG | : Not regulated as a dangerous good |
| IATA | : Not regulated as a dangerous good |

#### 14.4 Packing group

|                  |                                     |
|------------------|-------------------------------------|
| ADR              | : Not regulated as a dangerous good |
| RID              | : Not regulated as a dangerous good |
| IMDG             | : Not regulated as a dangerous good |
| IATA (Cargo)     | : Not regulated as a dangerous good |
| IATA (Passenger) | : Not regulated as a dangerous good |

#### 14.5 Environmental hazards

|                  |                                     |
|------------------|-------------------------------------|
| ADR              | : Not regulated as a dangerous good |
| RID              | : Not regulated as a dangerous good |
| IMDG             | : Not regulated as a dangerous good |
| IATA (Passenger) | : Not regulated as a dangerous good |
| IATA (Cargo)     | : Not regulated as a dangerous good |

#### 14.6 Special precautions for user

Not applicable

#### 14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

|         |                                           |
|---------|-------------------------------------------|
| Remarks | : Not applicable for product as supplied. |
|---------|-------------------------------------------|

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### SECTION 15: Regulatory information

#### 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

|                                                                                                                                                                           |   |                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------|
| REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59).                                                                                 | : | This product does not contain substances of very high concern (Regulation (EC) No 1907/2006 (REACH), Article 57).                |
| REACH - List of substances subject to authorisation (Annex XIV)                                                                                                           | : | Not applicable                                                                                                                   |
| Regulation (EC) No 1005/2009 on substances that deplete the ozone layer                                                                                                   | : | Not applicable                                                                                                                   |
| Regulation (EU) 2019/1021 on persistent organic pollutants (recast)                                                                                                       | : | Not applicable                                                                                                                   |
| Regulation (EC) No 649/2012 of the European Parliament and the Council concerning the export and import of dangerous chemicals                                            | : | Not applicable                                                                                                                   |
| REACH - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, preparations and articles (Annex XVII)                            | : | Not applicable                                                                                                                   |
| Seveso III: Directive 2012/18/EU of the European Parliament and of the Council on the control of major-accident hazards involving dangerous substances.<br>Not applicable |   |                                                                                                                                  |
| Volatile organic compounds                                                                                                                                                | : | Directive 2010/75/EU of 24 November 2010 on industrial emissions (integrated pollution prevention and control)<br>Not applicable |

#### 15.2 Chemical safety assessment

This information is not available.

### SECTION 16: Other information

#### Full text of H-Statements

|        |   |                                                                          |
|--------|---|--------------------------------------------------------------------------|
| H315   | : | Causes skin irritation.                                                  |
| H361fd | : | Suspected of damaging fertility. Suspected of damaging the unborn child. |
| H412   | : | Harmful to aquatic life with long lasting effects.                       |
| H413   | : | May cause long lasting harmful effects to aquatic life.                  |

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### Full text of other abbreviations

GB EH40 : UK. EH40 WEL - Workplace Exposure Limits  
GB EH40 / TWA : Long-term exposure limit (8-hour TWA reference period)

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road; AIIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

### Further information

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